3.1.1 Atomic structure: Sheet 3

Isotopes and mass spectrometry · 19 marks · about 25 minutes

Key information

- **A_r** = Σ(m/z × abundance) ÷ total abundance, compared with 1/12 of the mass of a ^{12}C atom.
- Electron impact: $X(g) + e^- \rightarrow X^+(g) + 2e^-$. Electrospray: $X(g) + H^+ \rightarrow XH^+(g)$, so subtract 1 from the m/z to get M_r.
- All ions get the same kinetic energy, so lighter ions travel faster and reach the detector first.
- **KE = ½mv²** and **v = d ÷ t**. Mass of one ion in kg = (mass of 1 mol in g ÷ 1000) ÷ L.
- Ions gain an electron at the detector, creating a current; abundance ∝ size of the current.

Section A: Self-quiz

01 Define the term relative atomic mass.

[2 marks]

02 A TOF mass spectrum was obtained for a sample of barium that contains the isotopes ^{136}Ba, ^{137}Ba and ^{138}Ba. The sample was ionised by electron impact. Identify the ion with the longest time of flight.

[1 mark]

03 Which row shows the order of discovery, from earliest to latest?

[1 mark]

A electron, neutron, nucleus
B electron, nucleus, neutron
C nucleus, electron, neutron
D nucleus, neutron, electron

04 JJ Thomson suggested that negative electrons were found throughout an atom like 'plums in a pudding of positive charge'. State two differences between the 'plum pudding' model and the model of atomic structure used today.

[2 marks]

05 Which atom has the smallest number of neutrons?

[1 mark]

A ^3H
B ^4He
C ^5He
D ^4Li

Section B: Exam questions

06 A sample of krypton is ionised using electron impact. The mass spectrum of this sample of krypton has four peaks.

Table 1

m/z	82	83	84	86
Relative intensity	6	1	28	8

06.1 Calculate the relative atomic mass (A_r) of this sample of krypton.
Give your answer to 1 decimal place.

[2 marks]

06.2 In a TOF mass spectrometer, each $^{84}Kr^{+}$ ion is accelerated to a kinetic energy of 4.83×10^{-16} J and the time of flight is 1.72×10^{-5} s

The kinetic energy of an ion is given by the equation $KE = \frac{1}{2}mv^2$ (KE / J, m / kg, v / m s^{-1})

Calculate the length, in metres, of the TOF flight tube.

The Avogadro constant, $L = 6.022 \times 10^{23}$ mol^{-1}

[4 marks]

07 A sample of strontium has a relative atomic mass of 87.7 and consists of three isotopes, ^{86}Sr, ^{87}Sr and ^{88}Sr

In this sample, the ratio of abundances of the isotopes ^{86}Sr : ^{87}Sr is 1 : 1

State why the isotopes of strontium have identical chemical properties.

Calculate the percentage abundance of the ^{88}Sr isotope in this sample.

[4 marks]

08 Explain how ions are detected and relative abundance is measured in a TOF mass spectrometer.

[2 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	M1 The average mass of an atom of an element / (weighted) average mass of all isotopes of an element M2 compared to 1/12th the mass of an atom of carbon-12	Source: 7404/1 Nov 2020 Q03.1	2
02	$^{138}Ba^{+}$	Source: 7405/1 Jun 2018 Q04.7	1
03	**B**	Source: 7404/1 Jun 2025 Q12	1
04	Statement about the nucleus: (central) nucleus contains protons and neutrons Statement about electrons: electrons are now arranged in energy levels / shells / orbitals	Source: 7404/1 Jun 2018 Q01.1 Allow protons and neutrons are in the centre of the atom Ignore 'mostly empty space'; ignore electrons surround / orbit nucleus	2
05	**D**	Source: 7404/1 Nov 2020 Q09	1
06.1	M1 [(82 × 6) + (83 × 1) + (84 × 28) + (86 × 8)] ÷ 43 = 3615 ÷ 43 M2 = 84.1	Source: 7404/1 Nov 2020 Q03.2–03.3 36.2 scores 1/2	2
06.2	M1 m = (84 ÷ 1000) ÷ 6.022×10^{23} (= 1.395×10^{-25} kg) M2 $v^2 = 2KE \div m = 2 \times 4.83 \times 10^{-16} \div 1.395 \times 10^{-25}$ M3 v = √(6 924 731 183) = 83 215 m s^{-1} M4 d = v × t = 83 215 × 1.72×10^{-5} = 1.43 (m)	Allow answers in range 1.43 – 1.44 m If m not converted to kg, d = 0.045 m for max 3	4
07	M1 Same electronic configuration / same number of electrons (in outer shell) / all have 37 electrons M2 86x + 87x + 88(100 – 2x) = 87.7 × 100 M3 x = 10% M4 % abundance of ^{88}Sr = 100 – 2 × 10 = 80(.0)%	Source: 7405/1 Jun 2018 Q04.6 Ignore protons and neutrons unless incorrect numbers Allow other alternative methods	4
08	M1 ion hits the detector / negative plate and gains an electron M2 (relative) abundance is proportional to (the size of) the current	Source: 7404/1 Jun 2018 Q08.2 Not positive plate	2